

Module 4

The Time Value of Money

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)

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 - Financial calculators

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 - Financial calculators
- **Part 3**
 - Special cases: perpetuities, annuities, and their variants

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 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
 - Financial calculators
- **Part 3**
 - Special cases: perpetuities, annuities, and their variants
- **Part 4**
 - The role of inflation

Module 4, Part 1

Future Value and Present Value

The Time Value of Money



Would you rather receive a dollar today, or a dollar tomorrow?

The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!
 - Assuming, there is some investment with a positive return for us to pursue

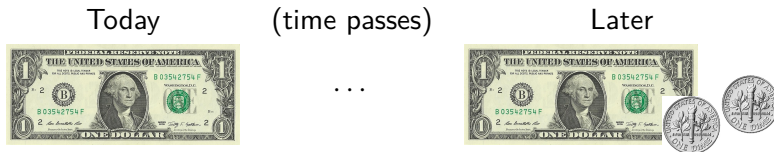


Future Value

- **Future value**, the amount to which an investment will grow after earning interest
- A “present value” today grows to its “future value” tomorrow via interest

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$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

Compound Interest

- **Compound interest**, the interest you earn with compounding, that is, with interest on interest

Future Value, Compound Interest

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Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

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- With compound interest, your \$100 grows to $\$100(1 + r)^T = 133.82$, so you earned \$33.82 in compound interest

Compound Interest

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Compound Interest

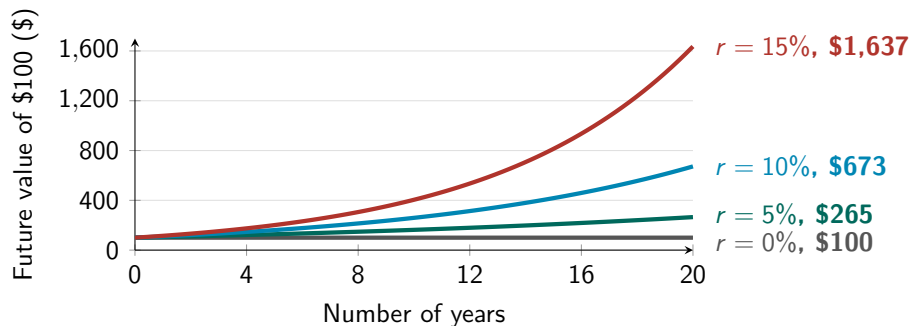
- **Compound interest**, the interest you earn with compounding, that is, with interest on interest
 - If you earn r per year for T years, your money grows to $PV \times (1 + r)^T$
 - PV of that total is the principal, so the total compound interest you have earned is $PV \times (1 + r)^T - PV$

Compound Interest

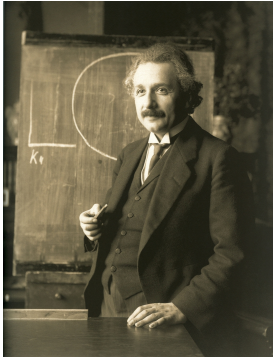
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- Compound interest is the default in finance (e.g., deposit rate)

Future Value Grows Fast in r and T

- The future value (FV) depends heavily on the interest rate r , especially as the time dimension T grows

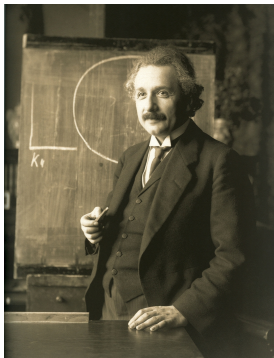


What's the Most Powerful Force in the Universe?



- Einstein, ...

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- Einstein, ...
- “Compound interest”

Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?



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(roughly the same as total world wealth)

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b. $FV = \$24 \times (1 + 0.035)^{395} = \19 million

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How many years will it take until your money doubles?



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- 11.896 years, because $(1 + 0.06)^{11.896} = 2$
 - We will learn how to calculate this with the financial calculator



Present Value

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- Recall, Future value $(FV) = PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T}$$

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- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value

Present Value

- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**

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- **Discount factor**, the present value of \$1 received in year T when the discount rate is r
 - Less than one when r is positive

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 **Discount factor**

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 **Discount factor**

- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

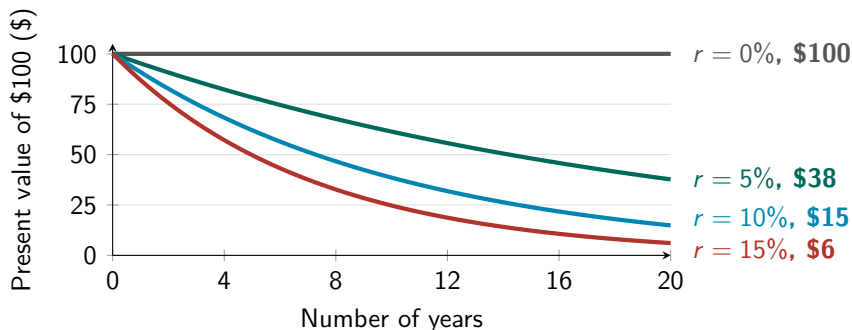
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^T} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

Present Value

- The present value (PV) of \$100 in the future depends heavily on the discount rate r , especially as the time dimension T grows



Present Value, Multiple Future Cash Flows

- Present values can be added together to calculate the total PV of multiple future cash flows

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$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

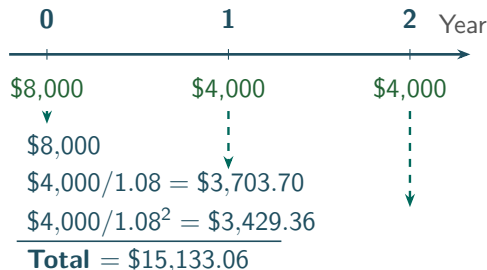
- where C_1 is the cash flow in one year, C_2 is the cash flow in two years, etc.

Example, Multiple Future Cash Flows

- You face two investments with different cash flows in the future
 - A. Receive \$15,000 cash now, or
 - B. Receive three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

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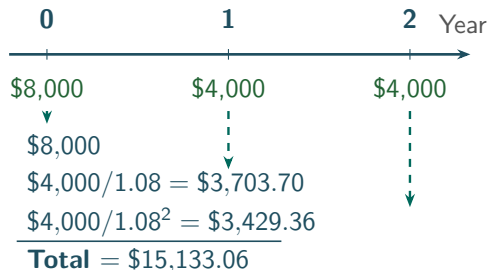


$$PV = \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2}$$

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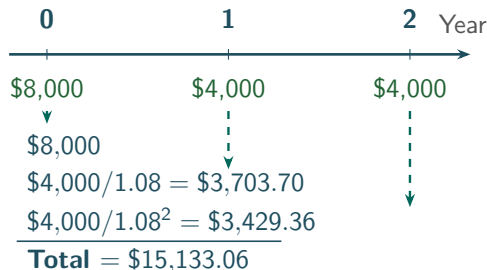
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0	1	2	Year
\$8,000	\$4,000	\$4,000	
▼	⋮	⋮	
\$8,000	\$4,000/1.08 = \$3,703.70		
	\$4,000/1.08 ² = \$3,429.36		
Total = \$15,133.06			

- The present value of option B (\$15,133.06) is more than option A's \$15,000, so choose option B
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

Return

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- Suppose Oracle's stock price was \$100 at the beginning of 2024 and \$180 at the end of 2024. What was the return for 2024?

$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

Holding Period Return

- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \dots, r_T$ gives

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$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 1.80 \times 0.80 - 1 = 44\%$$

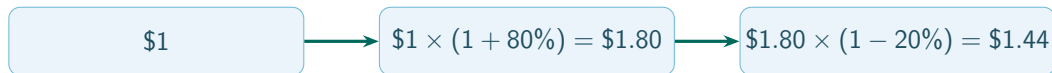
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- The annualized return is *not* $\frac{80\% - 20\%}{2} = 30\%$

Recap, Three Kinds of Returns

Return

Percentage growth of an asset's value over some period of time

Holding period return

Percentage growth over the whole period an asset is owned, longer or shorter than a year

Annualized return

Percentage growth per year, a geometric mean over the T years of the holding period

Recap, Three Kinds of Returns

Return	Holding period return	Annualized return
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

- These financial returns are *not directly related in any way* with **Module 3's** accounting-based ROA and ROE measures

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow

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- **Future value**, the amount to which an investment will grow after earning interest

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Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
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$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

- **Present value**, the value of a future cash flow multiplied by its discount factor

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

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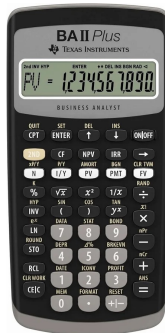
- Present values can be added together to calculate the total present value of multiple future cash flows

Module 4, Part 2

Financial Calculators

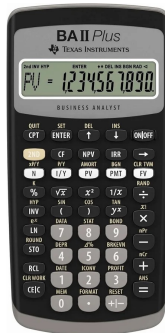
Get the Texas Instruments BA II Plus

- You need a **financial calculator**



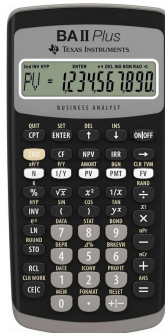
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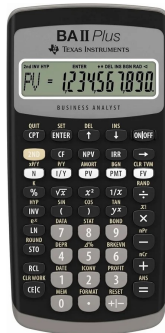
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- CFA (Chartered Financial Analyst), a key certification, often required in finance jobs
- The CFA exams allow only two calculator series, and the **TI BA II Plus** is one of them



Reset Your Financial Calculator

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- After you reset, you should repeat the next two settings!

Set the Number of Significant Digits to at Least 4!

- On the TI BA II Plus, press
 - 2nd
 - .
 - 4
 - ENTER

Set the Number of Compounding Periods per Year to 1!

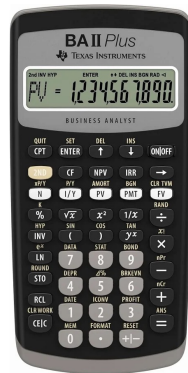
- On the TI BA II Plus, press
 - 2nd
 - I/Y (P/Y)
 - 1
 - ENTER

Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
 - 2nd
 - I/Y (P/Y)
 - 1
 - ENTER
- You must do this!

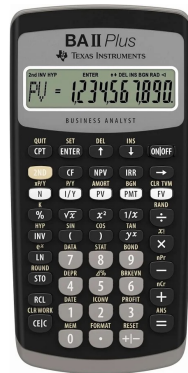
Financial Calculators, Basic Functionality

- N, number of periods (usually years)



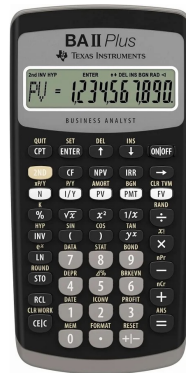
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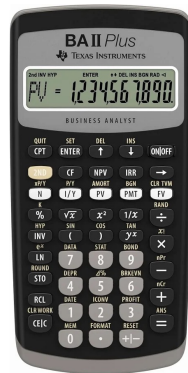
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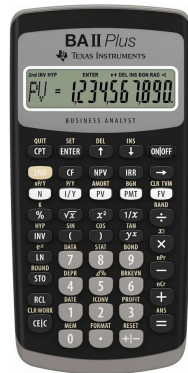
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- Why does the answer come out negative?

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- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
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- Bonus, what would the payment be if the monthly interest was 0.2%?

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- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest rate is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?

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- Let's try to solve some of the previous examples using Excel!

Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
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2015	21	
2016	20	
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Total	265	10

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- Suppose the discount rate is 10%. What is the present value of his contract?



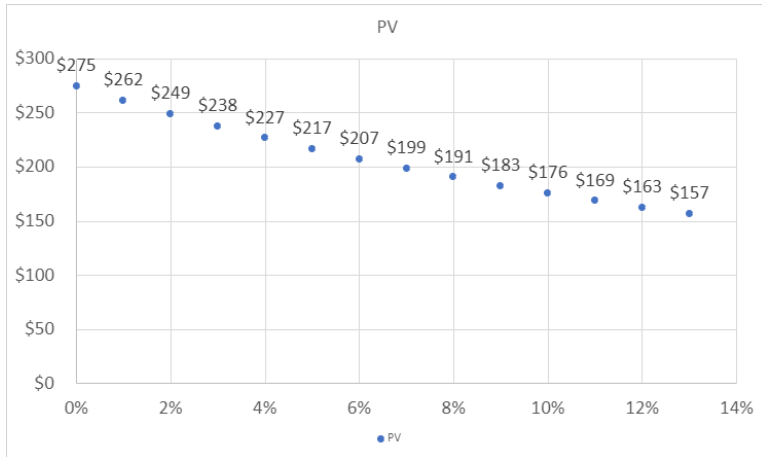
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- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?
- Why might it be useful for A-Rod to know the PV, and not simply the (undiscounted) sum of all future payments?



The PV of Payments to A-Rod



Recap of Part 2

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- Set the number of compounding periods per year to 1, and show at least 4 significant digits
- Excel has the same functionality, with the rate in decimals

Module 4, Part 3

Special Cases: Perpetuities and Annuities

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- Why care about perpetuities and annuities?
 - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)
 - There exist neat shortcuts that make it very easy to calculate the present value of these patterns of cash flows

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



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- C is the annual cash flow (aka the payment), and r is the discount rate

Example, Perpetuities

- Recall A-Rod's Yankees contract. Suppose the discount rate is 10%, then the PV of the contract (paying \$275m over 10 years) is \$175.6m
- Now suppose the Yankees offer A-Rod two options,
 - A. The contract as previously described (10 years with \$275m in total), or
 - B. A perpetual payment of C dollars every year. He will play for 10 years, but continue to be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- How much should the Yankees offer as the perpetual payment C for him to be **indifferent** between contract (A) and the “being paid forever” contract (B)?



Example, Perpetuities

- Solution, he is indifferent if the contracts have the same present value!

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$$PV_{\text{Contract A}} = PV_{\text{Contract B}} = \frac{C_B}{r} \Rightarrow C_B = PV \times r = 175.6 \times 10\% = \$17.56m$$

Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



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- **Delayed perpetuity**, first payment at time $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity

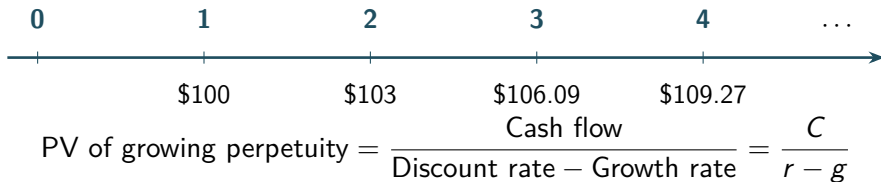
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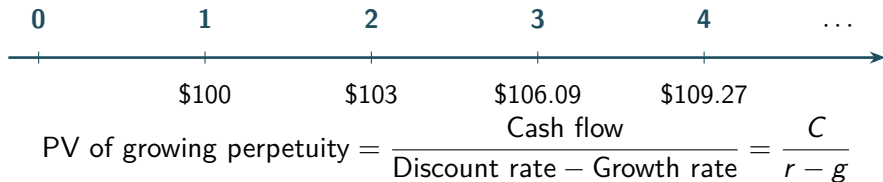
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- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Modules 5 and 9](#))
- When $g = 0$, the PV simplifies to the formula for a regular (non-growing) perpetuity

Annuities

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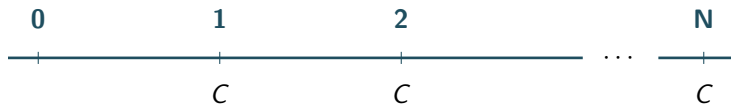


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- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)}_{\text{present value annuity factor (PVAF)}}$$

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- C is the annual cash flow (aka the payment), r is the discount rate, and N is the number of years that the annuity lasts

Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

Example, Annuity

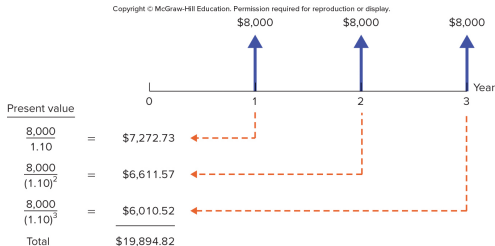
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$$PV = C \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left(1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

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Example, Annuity

- Suppose the discount rate for A-Rod's contract is 10%, so the present value is \$175.6m
- Now suppose instead the Yankees offer an annuity for 10 years with the same present value
- How much would he get paid per year, and in total (undiscounted) over 10 years, under such a same-value annuity contract?



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$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

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- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

- So he would be paid \$28.58m per year, or \$285.8m in undiscounted total over 10 years (slightly more in total, because his actual contract is front-loaded, which raises its present value compared to a constant payment)

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

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- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



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- **Annuity due**, first payment at time 0 $PV = PV \text{ annuity} \times (1 + r)$



- **Ordinary annuity**, first payment at time 1 $PV = PV \text{ annuity}$



- **Delayed annuity**, first payment at time $t > 1$ $PV = PV \text{ annuity} \times \frac{1}{(1+r)^{t-1}}$



Practice Problem

- A store offers two payment plans. Under the installment plan, you pay 25% down and 25% of the purchase price in each of the next 3 years. If you pay the entire bill immediately, you can get a discount of 7% on the purchase price. Assume the product sells for \$100
- a-1. Calculate the present value of the payments if you can borrow or lend funds at an interest rate of 4%
- a-2. Which is a better deal?
- b-1. Calculate the present value if the payments on the 4-year installment plan do not start for a full year
- b-2. Which is a better deal?

Practice Problem

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- a-2. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator, $PMT = -25$, $N = 3$, $FV = 0$, $I/Y = 4$, leading to 69.38, and next, $69.38 + 25 = 94.38$

Practice Problem

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- On a financial calculator, $PMT = -25$, $N = 4$, $FV = 0$, $I/Y = 4$

Practice Problem

- A local bank advertises the following deal, “Pay us \$100 at the end of each year for 11 years, and then we will pay you (or your beneficiaries) \$100 at the end of each year forever”
 - a. Calculate the present value of your payments to the bank if the interest rate is 7.00%
 - b. What is the present value of a \$100 perpetuity deferred for 11 years if the interest rate is 7.00%?
 - c. Is this a good deal?

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

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$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

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- On a calculator, PMT = 100, I/Y = 7, N = 11, FV = 0. Find PV

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- b. The bank's payments to you are a perpetuity deferred for 11 years

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$$PV_{11} = \frac{C}{r}$$
$$PV_{11} = \frac{100}{7\%} = \$1,428.57$$
$$PV = \frac{PV_{11}}{(1+r)^{11}} = \frac{1,428.57}{(1+7\%)^{11}} = \$678.70$$

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- c. Not a good deal (you pay \$749.83 for something worth \$678.70)

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$$\text{PV of growing annuity} = \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right]$$

Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
 - A. A lump sum of \$327 million, or
 - B. 30 installments of a growing annuity due with $g = 4\%$, starting at \$9.43m, with 30 payments, totaling around \$529 million



Photo, AP/Mark Humphrey

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- Which option should they take?



Photo, AP/Mark Humphrey

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

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- Suppose the discount rate is 3%

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...	...
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Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$

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$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$

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$$\begin{aligned} & \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^N \right] \times (1+r) \\ &= \frac{9.43}{0.03-0.04} \left[1 - \left(\frac{1+0.04}{1+0.03} \right)^{30} \right] \times (1+0.03) \\ &= \$326.58\text{m} \end{aligned}$$

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- That is, the present value of the annuity is almost exactly the same as (but slightly lower than) the \$327m lump sum

Practice Problem

- Professor's Annuity Corporation offers a lifetime annuity to retiring professors. For a payment of \$79,000 at age 65, the firm will pay the retiring professor \$575 a month until his death
- a. If the professor's remaining life expectancy is 20 years, what is the monthly interest rate on this annuity?
- b. If the monthly interest rate is 1.00%, what monthly annuity payment can the firm offer to the retiring professor?

Practice Problem

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$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $PMT = 575$, $N = 240$, $PV = -79,000$, $FV = 0$. Find I/Y

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- On a calculator, $\text{PMT} = 575$, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$. Find I/Y
- b.

$$C \times \left[\frac{1}{0.01} - \frac{1}{0.01 \times (1.01)^{240}} \right] = \$79,000 \Rightarrow C = \text{PMT} = \$869.86$$

- On a calculator, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$, $\text{I/Y} = 1$. Find PMT

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- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)
- These have handy formulas for calculating their present values!
- Timing,
 - Ordinary perpetuities and annuities make their first payment at time $t = 1$
 - Perpetuity and annuity dues make their first payment at $t = 0$
 - Delayed perpetuities and annuities make their first payment at $t > 1$

Module 4, Part 4

Inflation

History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
Source: U.S. Bureau of Labor Statistics.

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- **Nominal interest rate or nominal return**, the rate at which money invested grows
 - Compare dollars at time $t = 0$ to dollars at time $t = 1$
- **Real interest rate or real return**, the rate at which purchasing power increases
 - Compare the ability to buy a basket of goods at time $t = 0$ to the ability to buy the same basket of goods at time $t = 1$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

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Approximation, $\text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$

- Inflation also affects discounting,
 - We should discount nominal dollar cash flows using a nominal discount rate
 - We should discount real dollar cash flows using a real discount rate

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- Approximation, $\text{real} \approx 6\% - 2\% = 4.0\%$

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- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

Thank you!

Please let me know if you have any questions.